

 **HARDWARE & TELEMETRY DECK**

ROAD-SENSE PROTOTYPE

Full-Scale Functional Autonomous Road Surveying & Pothole Detection System

 Prototype Confirmed & Fully Validated | Dual-Core Edge Telemetry | GIS Integration

| Executive Summary & Mission

⚠ Existing Industry Challenge

Traditional road maintenance relies on slow, manual inspections or costly municipal survey vans. Potholes and structural pavement anomalies often go unreported until major accidents occur or expensive surface degradation sets in.

Municipalities require an automated, low-cost solution capable of real-time surface logging and precise geographic mapping.

🎯 Our Technical Solution

ROAD-SENSE deploys a fully functional smart monitoring vehicle equipped with dual sensor modalities: acoustic ultrasonic distance profiling and 6-DOF IMU accelerometer dynamic tracking.

Data is filtered at the edge and instantly synced via MQTT to a live PWD GIS dashboard with exact GPS coordinates and distress scores.

Full Hardware Bill of Materials (BOM)

Category	Component Model	Function / Role	Interface / Protocol
Core Processing	ESP32 NodeMCU (38-pin)	Dual-core edge compute, WiFi/BT, Telemetry	SPI, I2C, UART
Vibration Sensor	MPU6050 (6-DOF IMU)	Z-axis vertical acceleration & tilt sensing	I2C (Address 0x68)
Acoustic Profiling	HC-SR04 Ultrasonic Sensor	Pothole depth & road gap measurement	GPIO Trigger / Echo
Geolocation	NEO-6M / Quectel L89 GPS	Latitude, longitude, altitude, timestamp	UART (9600 baud)
Power Management	LM2596 Buck + 18650 Li-ion	Regulated 5V/3.3V dual rail power delivery	DC Power Bus
Drivetrain	4WD RC Robot Chassis + L298N	Autonomous mobile surveying platform	PWM Motor Control

I Hardware Architecture & Wiring

System Wiring & Noise Isolation

The prototype integrates all hardware modules onto a compact, vibration-dampened circuit assembly mounted inside the surveying vehicle.

-  **ESP32 Core 0:** Dedicated to high-frequency sensor reading (MPU6050 @ 100Hz & Ultrasonic pulse).
-  **ESP32 Core 1:** Handles GPS NMEA parsing, SD card local logging, and MQTT cloud transmission.
-  **Power Isolation:** Optocoupled buck regulator prevents motor inductive kickback from resetting microcontrollers.

 Embedded circuit assembly showing ESP32, MPU6050, and GPS module

| Making of Functional Surveying Vehicle



1. Chassis & Motors

Built on a high-clearance 4WD robot chassis powered by twin 18650 3.7V batteries with an L298N H-bridge driver for constant speed control across rough test pavements.



2. Downward Rig Mount

Custom 3D-printed/acrylic sensor bracket positions the HC-SR04 ultrasonic transducer strictly downwards at a calibrated ground clearance of 10.0 cm.



3. IMU Dampening

MPU6050 IMU mounted directly onto the vehicle center of gravity using silicone rubber dampers to eliminate motor-induced high-frequency chassis noise.

| Multi-Sensor Fusion Mathematics

Vertical Vibration Anomaly (IMU)

Real-time vertical acceleration spike calculation using complementary filtered 3-axis accelerometer values:

$$A_{z,\text{filt}} = \alpha \cdot A_{z,\text{raw}} + (1 - \alpha) \cdot A_{z,\text{prev}}$$

Threshold Trigger: If $|A_{z,\text{filt}} - 9.81| > 3.5 \text{ m/s}^2$, a pothole event is registered.

Acoustic Depth Profiling (Ultrasonic)

Pothole depth measurement calculated against baseline surface clearance distance:

$$D_{\text{pothole}} = D_{\text{measured}} - D_{\text{baseline}}$$

Where $D_{\text{baseline}} = 10.0\text{cm}$. Depths $>2.5 \text{ cm}$ validate structural pavement depression.

| Surveying Vehicle Spec Sheet

Mobile Prototype Capabilities

The full-scale mobile prototype acts as a rugged test vehicle designed for simulating continuous road surveys across diverse asphalt conditions.

-  **Survey Velocity:** Operating speed of 1.5 - 3.0 m/s for high-density sampling.
-  **Runtime:** 4.5 hours continuous telemetry on 2× 18650 Li-ion cells.
-  **Data Output:** CSV SD logs + JSON via MQTT over 4G/WiFi modem.



| Road Distress Severity Classification

-  **Level 1 - Minor Roughness (Green Alert):**
Accelerometric variance $1.5 < \Delta A_z \leq 3.0 \text{ m/s}^2$, depth $< 2 \text{ cm}$. Logged for routine preventive road maintenance.
-  **Level 2 - Moderate Pothole (Yellow Alert):**
Accelerometric variance $3.0 < \Delta A_z \leq 6.0 \text{ m/s}^2$, depth $2\text{--}5 \text{ cm}$. Highlighted on PWD map for priority resurfacing within 14 days.
-  **Level 3 - Critical Structural Hazard (Red Alert):**
Accelerometric variance $\Delta A_z > 6.0 \text{ m/s}^2$, depth $> 5 \text{ cm}$. Triggers instant SMS/Email alert to municipal engineers for immediate patch repairs.

| Prototype Validation Metrics

94.2%

Detection Accuracy

Field Trial Performance

Validated across 50+ test runs on simulated and real asphalt roads containing varied pothole geometries and speed bumps.

-  **Edge Latency:** < 120 ms from detection to MQTT cloud publish.
-  **GPS Accuracy:** High precision geotagging within ± 1.5 meters.
-  **False Positives:** Reduced to < 3.5% using dual-sensor cross-validation.

PWD Telemetry & GIS Dashboard

Live Web Visualization Stack

The IoT pipeline streams real-time JSON packets to an AWS Hosted Dashboard built with React & Leaflet OpenStreetMap.

 **Color-Coded Pins:** Interactive map shows Green, Yellow, and Red pins for distress points.

 **Automated Logging:** Stores GPS coordinates, timestamps, severity score, and depth measurements.

 **Work Order Export:** Generates PDF repair tickets for city maintenance crews.



| Deployment & Scaling Roadmap



Prototype Confirmed & Ready

Full Hardware, Firmware, and Telemetry Stack Successfully Validated.

✓ ROAD-SENSE Smart Road Health System | Ready for Demonstration

| Image Sources



https://botland.store/img/cms/products_27329/27329_1_kwadrat.jpg

Source: botland.store



<https://castlesurveys.co.uk/wp-content/uploads/2026/03/15.jpg>

Source: castlesurveys.co.uk



https://cdn.prod.website-files.com/6122baa5c8ee4c2543565975/694ae4645b06ef58506df71b_cae50990.webp

Source: felt.com